

My Database of definitions

1 Games

Definitions: $\Gamma = (N, (A_i)_{i \in N}, (u_i)_{i \in N})$

def:game

A Game is a combinatorial problem involving *Players* N , *Strategies* A_i and *Utilities* u_i , where each *player* faces decision making using reasoning.

Definitions: $\sigma_i : A_i \rightarrow A_i$
 $S = (A_i)_{i \in N}$
 $S_i = A_i$
 $S_{-i} = (A_j)_{(j \neq i) \in N}$
 $S_\Pi = \prod_{i \in N} A_i$

Strategies σ corresponds to the *actions* available to each *player* among its *set of strategies* A_i , while a *Strategy profile* S refers to the set of *actions* chosen by all players. The set of S_Π encompasses all possible *strategy profiles*, and S_{-i} represents all *strategy profiles* excluding player i .

Definitions: $u_i : S_\Pi \rightarrow \mathbb{R}$
 $BR_i = \{S \in S_\Pi \mid u_i(S_i, S_{-i}) \geq u_i(S'_i, S_{-i}) \text{ for all } S'_i \in S_\Pi\}$
 $BR_i = \{S \in S_\Pi \mid u_i(S) \geq u_i(S') \text{ for all } S' \in S_\Pi, j \neq i : S'_j = S_j\}$
 $NE = \{S \in S_\Pi \mid S \in BR_i \text{ for all } i \in N\}$

The *Utility* u_i is a function that calculates a value representing the satisfaction or payoff for *player* i based on a *strategy profile* S . *Best Response* BR refers to a *strategy* that would be selected by a *player* to optimise its *utility* without affection the strategies chosen by others.

To identify an *Equilibrium* NE within a game, it's necessary to analyse whether a *strategy profile* represents a *best response* BR for each *player* $i \in N$. From a deterministic perspective, the equilibrium is calculated based on *pure strategies*, resulting in a *Pure Nash equilibrium*.

2 Constraint Games

These types of Games are modeled using a Constraint Satisfaction Problems (CSP) framework and solved with Constraint Technologies.

2.1 CSP

Definitions: $CSP = (X, (D_i)_{i \in X}, C)$

The definition of a CSP is formulated as a tuple $P = \langle X, D, C \rangle$ where X represents a set of variables $X = x_1, \dots, x_n$, D represents the set of correspondent domains of each variable $D = D_1, \dots, D_n$ such that $x_i \in D_i$ and C represents the set of constraints associated with the relations of those variables.

Definitions: $cpsolver : CSP \rightarrow \{(x_i, d_i) \mid x_i \in X, d_i \in D, \forall C_j \in C : C_j \text{ is satisfied}\}$

A *Constraint Solver* is a computational tool designed to find solutions for a constraint satisfaction problem *CSP*. A *cpsolver* aims to find assignments of values in D_i to variables X that satisfy all the given constraints C .

Definitions: $COP = (X, (D_i)_{i \in X}, C, f)$
 $f : \prod_{i \in X} D_i \rightarrow \mathbb{R}$

A constraint optimization problem *COP* extends the *CSP* framework by incorporating an objective function f , which is used to calculate the value to be optimised.

2.2 Game in constraint form

Definitions: $\Gamma^c = (N, X, (D_j)_{j \in X}, C)$
 $X = (X_i)_{i \in N} \cup X_E$
 $C = (C_i)_{i \in N} \cup C_H$

A *Constraint Game* Γ^c is defined as a tuple consisting of a set of *Players* N , a set of *variables* X composed of the union of variables controlled by each player $(X_i)_{i \in N}$ and additional variables X_E (existential variables), a set of *domains* D_j for each variable, and a set of *constraints* C that include objectives for each player $(C_i)_{i \in N}$ plus global constraints (hard constraints) C_H .

3 ω -regular Games

These types of Games are played on a directed graph called *Arena*, with a specific method for computing player utilities known as *Winning Conditions*, defined as follows:

3.1 Arena

Definitions: $G = (V, E, f)$

The Arena G is defined as a directed graph where vertices V are connected by edges E , and each vertex has an additional property mapped by the function f .

Definitions: $uE := \{v \in V \mid (u, v) \in E\} \neq \emptyset \text{ for all } u \in V$

Every vertex has at least one edge directed away from it. Therefore, we can infer that the graph must contain at least one cycle or infinite path.

Definitions: $f : V \rightarrow L$

The function f assigns labels or values from a finite set L to the vertices as colors.

3.2 Game in ω -regular form

Definitions: $\Gamma^\omega = (N, (V_i)_{i \in N}, (W_i)_{i \in N}, E, f)$

In a *omega-regular Game* Γ^ω , played over a graph G , each *Player* $i \in N$ governs a subset of vertices V_i , each *Player* has a set of *actions* A_i represented by their *Strategies* mapped over the edges E of the graph and a set W_i , representing their *Winning Conditions*. The function f assigns the color of the vertices.

Definitions: $\omega = \langle t_0, t_1, \dots, t_h, \dots \rangle$ for all $h \geq 0$, where $(t_k, t_{k+1}) \in E$ for all $k \geq 0$

A play ω corresponds to an infinite path in G , where the initial h states constitute a finite initial segment referred to as the *history*, preceding the infinite segment. Ω corresponds to every set of plays over G .

Definitions: $\sigma_i : V^*V_i \rightarrow V$

The function σ over a ω -regular game determines a *Pure Strategy* for *Player* i , assigning to each vertex $v \in V_i$ the next state $\sigma(v)$ such that $(v, \sigma(v)) \in E$. This *Strategy* is called *positional* when σ is calculated only taking account the current vertex, that is the case in *Memory-less Games*. A *Strategy Profile* $(\sigma_i)_{i \in \Pi}$ is a tuple consisting of every *Strategy* chosen by each *Player* i .

Definitions: $O_i = \{O_i \in \Omega \mid O_i = W_i \text{ for all } i \in N, \text{ where } (O_i \cap O_j) \neq \emptyset \text{ for all } (i \neq j) \in N\}$

The set O_i correspond to the *Objective* of *Player* i , it contains the set of paths or plays where *Player* i wins W_i . The *Objectives* of every *Player* are mutually exclusive.

3.3 Playing

In a *Multiplayer Game* a play ω is conducted in a turn-based manner using an unique token, what is moved from t_k to t_{k+1} by the *Player* i who controls the current vertex v , in accordance with its *set of strategies* A_i . It is helpful to specify a starting vertex (Γ^ω, v_0) .

In terms of applicability, we are focusing on *Two-player Turn-based* multiplayer games, where the objective of the *Game* Γ^ω is defined by the specifications of *Winning Conditions* W_i .

3.4 Winning conditions

Definitions: $Reach(M) = \{\langle t_0, t_1, \dots \rangle \in \Omega \mid t_k \in M \text{ for some } k \geq 0\}$
 $Safe(M) = \{\langle t_0, t_1, \dots \rangle \in \Omega \mid t_k \in M \text{ for all } k \geq 0\}$

Reachability condition requires that some states in the set $M \subseteq V$ be visited. Similarly, *Safety* condition requires that only the states in the set $M \subseteq V$ be visited.

Definitions: $Inf(\omega) = \{t \in \langle t_0, t_1, \dots \rangle \in \Omega \mid t_k = t \text{ for infinitely many } k \geq 0\}$
 $Büchi(M) = \{\omega \in \Omega \mid Inf(\omega) \cap M \neq \emptyset\}$
 $coBüchi(M) = \{\omega \in \Omega \mid Inf(\omega) \cap M = \emptyset\}$

Having a set vertices that are reached infinitely many times, $Inf(\omega)$, a *Büchi* condition requires that some states in the set $M \subseteq V$ be visited infinitely often. Conversely, the *coBüchi* condition requires that any state in the set $M \subseteq V$ be visited infinitely often.

Definitions: $Pairs = \{(B_1, F_1), \dots, (B_d, F_d)\}$ where $1 \leq j \leq d$
 $Rabin(Pairs) = \{\omega \in \Omega \mid \exists 1 \leq j \leq d (Inf(\omega) \cap B_j = \emptyset \wedge Inf(\omega) \cap F_j \neq \emptyset)\}$
 $Streett(Pairs) = \{\omega \in \Omega \mid \forall 1 \leq j \leq d (Inf(\omega) \cap B_j \neq \emptyset \vee Inf(\omega) \cap F_j = \emptyset)\}$

Given a set of vertex pairs (B, F) where $B \subseteq V$ and $F \subseteq V$, a *Rabin* condition requires that no vertex in B is visited infinitely often and at least one vertex in F is visited infinitely often. Contrarily, (in the context of $\neg p \vee q \equiv p \rightarrow q$) a *Streett* condition requires that if some vertex in B is visited infinitely often, then at least one vertex in F must be visited infinitely often.

Definitions: $priority : V \rightarrow \mathbb{N}_0$
 $Parity = \{\omega \in \Omega \mid \max\{priority(t) \mid t \in Inf(\omega)\} \text{ is even}\}$

Given a function *prior* that assigns a non-negative integer $\mathbb{N}_0$ label to each vertex, the *Parity* condition requires that the highest priority that is reached infinitely often must be *even*.

Definitions: $Müller(M) = \{\omega \in \Omega \mid Inf(\omega) \in M \text{ where } M \subseteq 2^S\}$

The *Müller* condition requires that the set of vertices visited infinitely often must be one of the sets of vertices specified in M .

3.5 Techniques for resolving

Definitions: $Attr_i(Z) = \{v \in V_i \mid E(v) \cap Z \neq \emptyset\} \cup \{v \in V_{-i} \mid E(v) \cap Z \subseteq Z\}$ for all $Z \subseteq V$ and $i \in N$

Given a set of vertex Z the attractor of Z refers to a set of vertex that player i can control the game to reach.

Definitions: $Attr_{\bigcirc}^0(U) = U$
 $Attr_{\bigcirc}^{n+1}(U) = Attr_{\bigcirc}^n(U) \cup \{v \in V_{\bigcirc} \mid \exists E(v) \in Attr_{\bigcirc}^n(U)\} \cup \{v \in V_{\square} \mid \forall E(v) \in Attr_{\bigcirc}^n(U)\}$
 $Attr_{\bigcirc}(U) = \bigcup_{i \geq 0} Attr_{\bigcirc}^i(U)$

3.6 Reduction to SAT

Definitions: $= (S_{v_0} \wedge \Phi_{\exists} \wedge \Phi_{\forall} \wedge \Phi_V \wedge \Phi_A)$
 $\Phi_{\exists} = \bigwedge_{v \in V_{\exists}} (S_v \rightarrow \bigvee_{w \in vE} T_{v,w}),$
 $\Phi_{\forall} = \bigwedge_{v \in V_{\forall}} (S_v \rightarrow \bigwedge_{w \in vE} T_{v,w}),$
 $\Phi_V = \bigwedge_{w \in V, w \neq v_0} ((\bigvee_{v \in Ew} T_{v,w}) \rightarrow S_w),$
 $\Phi_A = \bigwedge_{(v,w) \in E} (T_{v,w} \rightarrow \Psi_{v,w}),$

Definitions: $\Psi_{v,w} = \bigwedge_{p \in Odd^{<\Omega(w)}(G)} (x_p^v \geq x_p^w), \text{ if } \Omega(w) \text{ is even},$
 $\Psi_{v,w} = (x_{\Omega(w)}^v > x_{\Omega(w)}^w) \wedge \bigwedge_{p \in Odd^{<\Omega(w)}(G)} (x_p^v \geq x_p^w), \text{ if } \Omega(w) \text{ is odd},$

3.7 NOC

Definitions: $G = (V, E, \Omega)$, $\mathcal{CSP} = (\mathcal{X}, \mathcal{D}, \mathcal{C})$

$$\begin{aligned} \mathcal{C}_{v_0} &: \mathcal{V}_{v_0} = \top \\ \mathcal{C}_p &: \bigwedge_{v \in V_p} (\mathcal{V}_v \rightarrow (\sum_{w \in vE} \mathcal{E}_{(v,w)} = 1)) \\ \mathcal{C}_{\bar{p}} &: \bigwedge_{v \in V_{\bar{p}}} (\mathcal{V}_v \rightarrow \bigwedge_{w \in vE} \mathcal{E}_{(v,w)}) \\ \mathcal{C}_E &: \bigwedge_{w \in V, w \neq v_0} ((\bigvee_{v \in Ew} \mathcal{E}_{(v,w)}) \rightarrow \mathcal{V}_w) \\ \mathcal{C}_{\text{NOC}} &: \bigwedge_{v_0, \dots, v_k} ((\bigwedge_{i \in 0 \dots k-1} \mathcal{E}_{(v_i, v_{i+1})}) \rightarrow \bigwedge_{j \in 0 \dots k-1} (v_j = v_k \rightarrow \max\{\Omega(v_i) \mid i \in j \dots k\} \bmod 2 \neq \bar{p})) \end{aligned}$$

3.8 NOCQ

Definitions: $G = (V, E, \Omega, Q)$, $\mathcal{P} = (\mathcal{X}, \mathcal{D}, \mathcal{C})$

$$\begin{aligned} \mathcal{C}_{v_0} &: \mathcal{V}_{v_0} = \top \\ \mathcal{C}_p &: \bigwedge_{v \in V_p} (\mathcal{V}_v \rightarrow (\sum_{w \in vE} \mathcal{E}_{(v,w)} = 1)) \\ \mathcal{C}_{\bar{p}} &: \bigwedge_{v \in V_{\bar{p}}} (\mathcal{V}_v \rightarrow \bigwedge_{w \in vE} \mathcal{E}_{(v,w)}) \\ \mathcal{C}_E &: \bigwedge_{w \in V, w \neq v_0} ((\bigvee_{v \in Ew} \mathcal{E}_{(v,w)}) \rightarrow \mathcal{V}_w) \\ \mathcal{C}_{\text{NOC}}(f) &: \bigwedge_{v_0, \dots, v_k} \left(\left(\bigwedge_{i=0}^{k-1} \mathcal{E}_{(v_i, v_{i+1})} \right) \rightarrow \bigwedge_{j=0}^{k-1} ((v_j = v_k) \rightarrow f(v_j, \dots, v_k)) \right) \end{aligned}$$

Definitions: $f_\omega(v_j, \dots, v_k) = \max\{\Omega(v_i) \mid i \in j \dots k\} \bmod 2 \neq \bar{p}$

$$f_\eta(v_j, \dots, v_k) = \sum_{i=j}^{k-1} Q(v_i, v_{i+1}) \geq 0$$

$$f_\mu(v_j, \dots, v_k) = \frac{1}{k-j} \sum_{i=j}^{k-1} Q(v_i, v_{i+1}) \geq \nu$$

3.9 CrossNOCQ

Definitions: $G = (V, E, \Omega, Q)$, $\mathcal{CSP} = (\mathcal{X}, \mathcal{D}, \mathcal{C})$

The CSP encoding of the parity game with arena G , is defined as $\mathcal{CSP} = (\mathcal{X}, \mathcal{D}, \mathcal{C})$, where the set of variables is given by $\mathcal{X} = \{\mathcal{V}_v\}_{v \in V} \cup \{\mathcal{E}_e^\circ\}_{e \in E} \cup \{\mathcal{E}_e^\square\}_{e \in E}$. All of these variables are Boolean, thus, $\mathcal{D} = \{\perp, \top\}$. The set of constraints, denoted by $\mathcal{C}$, is defined as:

$$\begin{aligned} \mathcal{C}_p^\circ &: \bigwedge_{v \in V_p} (\mathcal{V}_v \rightarrow (\sum_{w \in vE} \mathcal{E}_{(v,w)}^\circ = 1)) \\ \mathcal{C}_{\bar{p}}^\circ &: \bigwedge_{v \in V_{\bar{p}}} (\mathcal{V}_v \rightarrow \bigwedge_{w \in vE} \mathcal{E}_{(v,w)}^\circ) \\ \mathcal{C}_E^\circ &: \bigwedge_{(u,w) \in E} (\mathcal{E}_{(u,w)}^\circ \rightarrow \mathcal{V}_w) \\ \mathcal{C}_{\text{NOC}}^\circ(f) &: \bigwedge_{C \in \text{cycle}(G)} \left(\left(\bigwedge_{e \in C} \mathcal{E}_e^\circ \right) \rightarrow f(C) \right) \\ \mathcal{C}_p^\square &: \bigwedge_{v \in V_p} (\neg \mathcal{V}_v \rightarrow (\sum_{w \in vE} \neg \mathcal{E}_{(v,w)}^\square = 1)) \\ \mathcal{C}_{\bar{p}}^\square &: \bigwedge_{v \in V_{\bar{p}}} (\neg \mathcal{V}_v \rightarrow \bigwedge_{w \in vE} \neg \mathcal{E}_{(v,w)}^\square) \\ \mathcal{C}_E^\square &: \bigwedge_{(u,w) \in E} (\neg \mathcal{E}_{(u,w)}^\square \rightarrow \neg \mathcal{V}_w) \\ \mathcal{C}_{\text{NOC}}^\square(f) &: \bigwedge_{C \in \text{cycle}(G)} \left(\left(\bigwedge_{e \in C} \mathcal{E}_e^\square \right) \rightarrow f(C) \right) \end{aligned}$$

A cycle in G is defined as $C = (e_0, e_1, \dots, e_{k-1})$ where $e_i = (v_i, v_{i+1})$ and $v_k = v_0$. $V(C)$ is the set of vertices in the cycle C .

Definitions: $f_\omega(C) = \max\{\Omega(v) \mid v \in V(C)\} \bmod 2 \neq \bar{p}$

$$f_\eta(C) = \sum_{e \in C} Q(e) \geq 0$$

$$f_\mu(C) = \frac{1}{|C|} \sum_{e \in C} Q(e) \geq \nu$$

3.10 MDPs

Definitions: $\mathcal{MDP} = (V_{\circ}, V_{\diamond}, E, R, \text{Pr})$, $\mathcal{CSP} = (\mathcal{X}, \mathcal{D}, \mathcal{C})$

An $\mathcal{MDP}$ is a finite set of vertices partitioned between Player $\circ$ and Nature $\diamond$, $E \subseteq V \times V$ is a set of directed edges where every vertex has at least one successor, $R : V_{\circ} \rightarrow \{0, \dots, d\}$ is a priority mapping, and $\text{Pr} : E_{\diamond} \rightarrow (0, 1]$ is a function assigning probabilities to edges originating from Nature vertices, where $E_{\diamond} = E \cap (V_{\diamond} \times V)$, satisfying $\forall v \in V_{\diamond} : \sum_{(v,u) \in E} \text{Pr}(v, u) = 1$.

Definitions: $\mathcal{C}_p^{\circ} : \bigwedge_{v \in V_p} (\mathcal{V}_v \rightarrow (\sum_{w \in vE} \mathcal{E}_{(v,w)}^{\circ} = 1))$
 $\mathcal{C}_{\bar{p}}^{\circ} : \bigwedge_{v \in V_{\bar{p}}} (\mathcal{V}_v \rightarrow \bigwedge_{w \in vE} \mathcal{E}_{(v,w)}^{\circ})$

3.11 Stochastic-CrossNOC

Definitions: $\mathcal{A} = (V_{\circ}, V_{\square}, V_{\diamond}, E, \text{Pr})$, $\mathcal{CSP} = (\mathcal{X}, \mathcal{D}, \mathcal{C})$

A game arena $\mathcal{A}$ is a finite set of vertices partitioned between Player $\circ$, Player $\square$ and Nature $\diamond$, $E \subseteq V \times V$ is a set of directed edges where every vertex has at least one successor, $\Omega : V \rightarrow \{0, \dots, d\}$ is a priority mapping, and $\text{Pr} : E_{\diamond} \rightarrow (0, 1]$ is a function assigning probabilities to edges originating from Nature vertices, where $E_{\diamond} = E \cap (V_{\diamond} \times V)$, satisfying $\forall v \in V_{\diamond} : \sum_{(v,u) \in E} \text{Pr}(v, u) = 1$.

The CSP encoding of the parity game with arena $\mathcal{A}$, is defined as $\mathcal{CSP} = (\mathcal{X}, \mathcal{D}, \mathcal{C})$, where the set of variables is given by $\mathcal{X} = \{\mathcal{V}_v\}_{v \in V} \cup \{\mathcal{E}_e^{\circ}\}_{e \in E} \cup \{\mathcal{E}_e^{\square}\}_{e \in E} \cup \{\mathcal{U}_e\}_{e \in E}$. All of these variables are Boolean, thus, $\mathcal{D} = \{\perp, \top\}$. The set of constraints, denoted by $\mathcal{C}$, is defined as:

Definitions: $\mathcal{C}_p^{\circ} : \bigwedge_{v \in V_p} (\mathcal{V}_v \rightarrow (\sum_{w \in vE} \mathcal{E}_{(v,w)}^{\circ} = 1))$
 $\mathcal{C}_{\bar{p}}^{\circ} : \bigwedge_{v \in V_{\bar{p}}} (\mathcal{V}_v \rightarrow \bigwedge_{w \in vE} \mathcal{E}_{(v,w)}^{\circ})$
 $\mathcal{C}_E^{\circ} : \bigwedge_{(u,w) \in E} (\mathcal{E}_{(u,w)}^{\circ} \rightarrow \mathcal{V}_w)$
 $\mathcal{C}_{\text{NOC}}^{\circ}(f) : \bigwedge_{C \in \text{cycle}(G)} ((\bigwedge_{e \in C} \mathcal{E}_e^{\circ}) \rightarrow f(C))$
 $\mathcal{C}_p^{\square} : \bigwedge_{v \in V_p} (\neg \mathcal{V}_v \rightarrow (\sum_{w \in vE} \neg \mathcal{E}_{(v,w)}^{\square} = 1))$
 $\mathcal{C}_{\bar{p}}^{\square} : \bigwedge_{v \in V_{\bar{p}}} (\neg \mathcal{V}_v \rightarrow \bigwedge_{w \in vE} \neg \mathcal{E}_{(v,w)}^{\square})$
 $\mathcal{C}_E^{\square} : \bigwedge_{(u,w) \in E} (\neg \mathcal{E}_{(u,w)}^{\square} \rightarrow \neg \mathcal{V}_w)$
 $\mathcal{C}_{\text{NOC}}^{\square}(f) : \bigwedge_{C \in \text{cycle}(G)} ((\bigwedge_{e \in C} \mathcal{E}_e^{\square}) \rightarrow f(C))$

A cycle in G is defined as $C = (e_0, e_1, \dots, e_{k-1})$ where $e_i = (v_i, v_{i+1})$ and $v_k = v_0$. $V(C)$ is the set of vertices in the cycle C .

Definitions: $f_{\omega}(C) = \max\{\Omega(v) \mid v \in V(C) \setminus V_{\diamond}\} \bmod 2 \neq \bar{p}$

4 Experimental Games

Still in progress ↓

4.1 Games in Enumeration Form

Definitions: $\Gamma^n = \{(S, (U_i)_{i \in N}) \mid S \in S_{\Pi}, U_i = u_i(S)\}$

Given a game Γ , a Γ^n corresponds to a set of pairs of tuples (S, U) , where S represents a *strategy profile*, and U is a tuple of utilities (payoffs) calculated with the function u_i for each *player* based on the *strategy profile* S .

4.2 Games in Constraint Form extended

Definitions: $\Gamma^c = (N, X, (D_j)_{j \in X}, C)$
 $X = (X_i)_{i \in N} \cup X_E$
 $C = (C_i)_{i \in N} \cup C_H \cup C_W$

4.3 Zero-sum two players ω -regular Games in Constraint Form

Definitions: $\Gamma^c = (N, (V_i)_{i \in N}, E, (C_i)_{i \in N}, f, v_0)$

Definitions: $N = \{\circ, \square\}$
 $V_\circ = \{v \in V \mid \text{owner}(v) = \circ\}$
 $V_\square = \{v \in V \mid \text{owner}(v) = \square\}$
 $E = \{(v, u) \mid v, u \in V\}$
 $f : V \rightarrow \mathbb{N}$
 $C_\circ = v_0 \wedge$
 $\quad \bigwedge_{v \in V_\circ} (V_v \rightarrow \bigvee_{u \in vE} E_{(v, u)}) \wedge$
 $\quad \bigwedge_{v \in V_\square} (V_v \rightarrow \bigwedge_{u \in vE} E_{(v, u)}) \wedge$
 $\quad \bigwedge_{v \in V} ((\bigvee_{u \in vE} E_{(v, u)}) \rightarrow V_v) \wedge$
 $\quad \bigwedge_{u \in V, u \neq v_0} ((\bigvee_{v \in Eu} E_{(v, u)}) \rightarrow V_u) \wedge$
 $\quad \{\omega \in \Omega \mid \min\{f(v) \mid v \in \text{Inf}(\omega)\} \text{ is even}\} \wedge$
 $\quad \{v_0, v_1, \dots\} \in \Omega \mid v_k \geq 0\}$
 $C_\square = \neg C_\circ$